

Digital Signal Processing — Formula Sheet

CSE 4631 · Quick reference: system properties, sampling, DFT/FFT, filter design, modulation

1 · System Properties

1.1 Linearity Test (Superposition)

$$T\{a \cdot x_1[n] + b \cdot x_2[n]\} = a \cdot T\{x_1[n]\} + b \cdot T\{x_2[n]\}$$

Compute the system's response to a weighted sum of two inputs, and separately compute the weighted sum of the two individual outputs. If they match for *all* a , b , x_1 , x_2 → linear.

Ex: $y[n] = 2x[n] + 3$. Combined-input response = $2ax_1 + 2bx_2 + 3$; weighted-sum-of-outputs = $2ax_1 + 2bx_2 + 3a + 3b$. These differ (the constant +3 doesn't scale) → **Non-Linear**. Any additive constant/offset breaks linearity.

1.2 Time-Invariance Test

$$T\{x[n-k]\} = y[n-k]$$

Feed a shifted input and compare to the correspondingly shifted original output. Match for all k → time-invariant (shift-invariant).

Ex: $y[n] = n \cdot x[n]$. Response to shifted input: $n \cdot x[n-k]$. Shifted output: $(n-k) \cdot x[n-k]$. Unequal → **Time-Variant** (any explicit dependence on absolute index n breaks time-invariance).

2 · Sampling, Nyquist & Aliasing

2.1 Nyquist Frequency

$$f_N = f_s / 2$$

Signals must be sampled at $f_s > 2 \cdot f_{\max}$ to avoid aliasing.

2.2 Aliased Frequency

$$f_{\text{alias}} = |f - k \cdot f_s|, \text{ chosen so } 0 \leq f_{\text{alias}} \leq f_s/2$$

Ex: $f = 700$ Hz sampled at $f_s = 1000$ Hz. $f_N = 500$ Hz, so aliasing occurs. With $k=1$: $f_{\text{alias}} = |700 - 1000| = 300$ Hz. The 700 Hz tone is indistinguishable from a 300 Hz tone once sampled — the lost information cannot be recovered by any digital filter.

3 · DFT Fundamentals

3.1 Real-Signal DFT Symmetry Rules

Component	Property	Requirement
Re $X[k]$	Even	Re $X[k] = \text{Re } X[N-k]$
Im $X[k]$	Odd	Im $X[k] = -\text{Im } X[N-k]$
DC bin ($k=0$)	Purely real	Im $X[0] = 0$
Nyquist ($k=N/2$)	Purely real	Im $X[N/2] = 0$

Ex (N=8): $X[0]=10+2j$ → invalid, Im $X[0] \neq 0$. $X[4]=3+1j$ (Nyquist, $N/2=4$) → invalid, Im $X[4] \neq 0$. $X[7]$ should equal conj($X[1]$)= $4+3j$ since $X[1]=4-3j$, but $X[7]$ is given as $4-3j$ → **fails odd-symmetry check**.

3.2 Frequency Resolution

$$\Delta f = f_s / N$$

N = number of time-domain samples. Longer records (bigger N) → finer resolution. Window shape (leakage) and SNR also limit practical resolution.

3.3 Frequency-Domain Interpolation

To upsample an N -point signal to M points ($M > N$): take the N -point DFT → zero-pad the *middle* (high-frequency end) of the spectrum with $(M-N)$ zeros → take the M -point IDFT. Adds no new frequency content, only increases time-domain sample density (sinc interpolation).

4 · Polar vs. Rectangular Notation

5 · FFT / DFT Computation

5.1 Bit-Reversal Index Mapping

For an N -point FFT, write index n in $k = \log_2 N$ bits, then reverse the bit order to get the new position.

Ex (N=16, k=4 bits): $n=3 \rightarrow 0011 \rightarrow \text{reversed } 1100 = 12$. $n=7 \rightarrow 0111 \rightarrow 1110 = 14$. $n=14 \rightarrow 1110 \rightarrow 0111 = 7$.

5.2 DFT vs. Radix-2 FFT Complexity

Direct DFT: $O(N^2)$ operations. Radix-2 FFT: $O(N \log_2 N)$ operations.

$$\text{Speedup } S = N^2 / (N \log_2 N) = N / \log_2 N$$

Ex (N=1024): DFT $\approx 1024^2 = 1,048,576$ ops. FFT $\approx 1024 \times 10 = 10,240$ ops. $S = 102.4 \times$ faster. (Fewer ops also means less accumulated round-off error → FFT is more accurate too.)

5.3 FFT Butterfly Unit

$$A = X_1 + W_N^k X_2 \quad B = X_1 - W_N^k X_2$$

Twiddle factor: $W_N^k = e^{-j2\pi k/N} = \cos(2\pi k/N) - j \sin(2\pi k/N)$. One complex multiply + one add + one subtract produce two outputs from two inputs — this reuse is what gives $O(N \log N)$ instead of $O(N^2)$. Combining two 8-point spectra $E(k)$, $O(k)$ into a 16-point spectrum: $X(k) = E(k) + W_{16}^k O(k)$, $X(k+8) = E(k) - W_{16}^k O(k)$, for $k=0 \dots 7$.

6 · Filter Design & Kernel Conversion

6.1 Windowed-Sinc FIR Design

$$\text{Unwindowed kernel: } h[n] = \frac{\sin(2\pi f_c(n-M/2))}{\pi(n-M/2)}$$

$$M \approx 4 / \text{BW} \quad (\text{BW, } f_c \text{ normalized to } f_s)$$

BW = transition-band width + f_s . f_c = cutoff + f_s . Larger $M \Rightarrow$ narrower transition band. (For a required stopband attenuation instead of a target M , use the window-specific constant: Rectangular $M \approx 0.9/\text{BW}$ [~ 21 dB], Hamming $M \approx 3.3/\text{BW}$ [~ 53 dB], Blackman $M \approx 5.5/\text{BW}$ [~ 74 dB] — pick the window that meets the dB spec.)

Ex: Remove noise above 4 kHz, $f_s=20$ kHz, transition BW=1 kHz. $\text{BW}_{\text{norm}} = 1/20 = 0.05$. $M = 4/0.05 = 80$ (use 81-pt symmetric kernel). $f_c = 4/20 = 0.20$.

6.2 Spectral Inversion (LP → HP, same cutoff)

$$h_{\text{HP}}[n] = \delta[n-M/2] - h_{\text{LP}}[n]$$

Negate every sample, then add 1 to the center-of-symmetry sample. Keeps the *same* cutoff frequency.

Ex: $h_{\text{LP}} = [-0.1, 0.2, 0.6, 0.2, -0.1]$. Negate → $[0.1, -0.2, -0.6, -0.2, 0.1]$; add 1 at center ($n=2$): $-0.6+1=0.4 \rightarrow h_{\text{HP}} = [0.1, -0.2, 0.4, -0.2, 0.1]$.

6.3 Spectral Reversal (frequency shift by π)

$$h_{\text{HP}}[n] = (-1)^n \cdot h_{\text{LP}}[n]$$

Flips every other sample's sign — equivalent to modulating by $0.5f_s$. Mirrors the response left-right about $f_s/4$, moving the cutoff from f_c to $(f_s/2 - f_c)$ and swapping pass/stop sides.

Ex: $h_{\text{LP}} = [0.05, 0.25, 0.40, 0.25, 0.05]$ → multiply by $(+1, -1, +1, -1, +1) \rightarrow h_{\text{HP}} = [0.05, -0.25, 0.40, -0.25, 0.05]$.

4.1 Magnitude, Phase & Quadrant Correction

$$\text{Mag } X[k] = \sqrt{(\text{Re } X[k]^2 + \text{Im } X[k]^2)}$$

$$\text{Phase } X[k] = \arctan(\text{Im } X[k] / \text{Re } X[k]) + \text{correction}$$

Standard arctan only returns $-90^\circ \dots +90^\circ$; when $\text{Re} < 0$ add/subtract 180° to land in the correct quadrant.

Ex 1: $\text{Re}=1, \text{Im}=1 \rightarrow \text{Mag}=\sqrt{2} \approx 1.414, \arctan(1)=45^\circ, \text{Re}>0$ so no correction $\rightarrow +45^\circ$.

Ex 2: $\text{Re}=-1, \text{Im}=-1 \rightarrow \text{Mag}=\sqrt{2} \approx 1.414, \text{uncorrected } \arctan(-1/-1)=45^\circ$, but $\text{Re}<0, \text{Im}<0$ (Quadrant III) $\rightarrow$ subtract $180^\circ \rightarrow -135^\circ$.

4.2 Polar-Notation Drawback: Phase Instability

When $\text{Mag } X[k] \approx 0$ (both $\text{Re}, \text{Im} \approx 0$), $\text{Im}/\text{Re} \rightarrow 0/0$ (indeterminate). Small background noise then makes the computed phase swing wildly between $-\pi$ and $+\pi$, so phase data in stopbands / near-zero-magnitude bins is unreliable. (Other issues: divide-by-zero when $\text{Re}=0$; 2π wraparound requiring "unwrapping"; more complex/non-Gaussian noise statistics than Re/Im .)

4.3 LTI Output From Polar Frequency-Response Data

$$\text{Re } X[k] = \text{Mag} \cdot \cos(\theta), \quad \text{Im } X[k] = \text{Mag} \cdot \sin(\theta)$$

Procedure: (1) convert both $X[k]$ and $H[k]$ to rectangular; (2) multiply $Y[k] = X[k] \cdot H[k]$ (complex multiplication); (3) take the IDFT of $Y[k]$ to get the time-domain output $y[n]$.

6.4 Band-Pass & Band-Reject Synthesis

$$h_{BP}[n] = h_{HP(\text{low-edge})}[n] * h_{LP(\text{high-edge})}[n] \quad (\text{cascade/convolve})$$

$$h_{BR}[n] = h_{LP(\text{low-edge})}[n] + h_{HP(\text{high-edge})}[n] \quad (\text{parallel/add})$$

Band-pass = *convolve* an HPF at the band's low edge with an LPF at the band's high edge (passes only what clears both). Band-reject = *add* an LPF at the low edge and an HPF at the high edge (passes everything outside the band). Use inversion/reversal above to obtain whichever cutoff/type you don't already have.

7 · Time-Domain / Frequency-Domain Duality

7.1 Phase Shift From a Time Shift

$$\Delta\text{Phase}[k] = -2\pi k \cdot s / N \quad (\text{radians, for a delay of } s \text{ samples})$$

Magnitude is unchanged by a pure time shift; phase changes *linearly* with frequency bin k — higher frequencies see a larger phase shift for the same time delay.

Ex: $N=128, s=2$ samples, $k=16$: $\Delta\text{Phase} = -2\pi(16)(2)/128 = -64\pi/128 = -\pi/2 \text{ rad } (-90^\circ)$.

8 · Convolution & Modulation

8.1 Discrete Linear Convolution

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

Output length = $\text{len}(x) + \text{len}(h) - 1$.

Ex: $x=[1,2,3], h=[1,-1]$. $y[0]=1 \cdot 1=1$; $y[1]=1(-1)+2(1)=1$; $y[2]=2(-1)+3(1)=1$; $y[3]=3(-1)=-3 \rightarrow \mathbf{y=[1,1,1,-3]}$.

8.2 Amplitude Modulation

$$y[n] = x[n] \cdot \cos(\omega_c n)$$

$$Y(\omega) = \frac{1}{2} X(\omega - \omega_c) + \frac{1}{2} X(\omega + \omega_c)$$

Time-domain multiplication by a carrier $\Leftrightarrow$ frequency-domain convolution with two impulses at $\pm\omega_c$. Splits the baseband spectrum into two symmetric copies (sidebands) centered at $+\omega_c$ and $-\omega_c$ — used to place signals at a convenient transmission frequency and to multiplex multiple signals on different carriers.